Rotation

Question 1 (Conceptual): Two discs, X and Y, with identical mass but different radii ($R_X = 0.5\,m, R_Y = 1.0\,m$), rotate about their axes with the same angular speed. Which disc has the greater rotational kinetic energy? Explain using $K_{rot} = \frac{1}{2} I w^2$.

Question 2 (Calculation): A solid sphere of mass 3.0 kg and radius 0.2 m rolls without slipping down a 5.0 m long incline that makes an angle of 30° with the horizontal.

(a) Write the relationship between translational and rotational kinetic energy for rolling without slipping.
(b) Calculate the sphere's speed at the bottom.
(c) Determine the total rotational kinetic energy at the bottom.